



Shell Scripting Advanced Assignment Lab – Uncovered Topics

Title: Advanced Practice & Evaluation Tasks (New Problem Set)

Purpose: Skill Expansion + Logical Thinking Assessment

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This assignment lab contains **new questions and problem statements** that were not directly covered in the training labs. These tasks are designed to evaluate deeper understanding, independent thinking, logic building, and real-world problem-solving skills.

Participants must write their own shell scripts for each task. Only instructions are provided. No sample code is given.

These tasks test the participant's ability to design automation logic independently.



Instructions for Participants

- Create a separate shell script for each task
 - Follow proper script structure (`#!/bin/bash`)
 - Use meaningful file names
 - Add execution permission to each script
 - Test each script before submission
 - Use comments to explain logic
 - Ensure scripts run without errors
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Advanced Assignment Tasks



Task 1: Dynamic Project Generator

Write a script that asks the user for a project name and automatically creates a complete project structure with multiple folders and files based on the input.



Task 2: Automated File Organizer

Create a script that scans a directory and automatically moves files into folders based on file type (for example: `.txt`, `.log`, `.sh`, `.conf`).

Task 3: Password Strength Validator

Write a script that takes a password as input and checks if it meets basic strength rules (length, numbers, special characters).

Task 4: User Activity Logger

Create a script that records login user information, date, time, and command history into a log file.

Task 5: Disk Cleanup Automation

Write a script that finds old files in a directory (based on days) and deletes them automatically.

Task 6: System Resource Alert

Create a script that monitors CPU, memory, and disk usage and prints alerts when usage crosses defined limits.

Task 7: Multi-User Environment Setup

Write a script that creates multiple users automatically from a text file input.

Task 8: Automated Configuration Backup

Create a script that takes backup of important configuration files and stores them with date and time stamp.

Task 9: Process Monitor

Write a script that checks if a specific process is running and logs its status continuously.

Task 10: Menu-Based Automation Tool

Create a script that shows a menu and allows the user to choose different system operations using numbers.

Task 11: Scheduled Task Generator

Write a script that automatically creates cron jobs based on user input.

Task 12: Log Analyzer

Create a script that reads a log file and extracts important information such as errors, warnings, and critical messages.

Task 13: Auto-Restart Service Tool

Write a script that continuously monitors a service and restarts it if it stops.

Task 14: System Report Generator

Create a script that generates a formatted system report file containing system info, user info, resource usage, and running services.

Task 15: Environment Setup Script

Write a script that prepares a development environment by installing tools, creating folders, and setting permissions.

Task 16: File Integrity Checker

Create a script that checks if important files are modified and alerts the user.

Task 17: Application Auto-Installer

Write a script that downloads and installs an application automatically.

Task 18: Backup Verification Tool

Create a script that verifies backup integrity and reports success or failure.

Task 19: Distributed System Simulation

Write a script that simulates managing multiple systems using loops and automation logic.

Task 20: Automation Toolkit Builder

Create a master script that combines multiple automation tools into one menu-based system.

Assessment Focus

This assignment evaluates: - Independent problem solving - Logical thinking - Automation design - Script architecture - Real-world scenario handling - Advanced system understanding - DevOps automation mindset

Submission Guidelines

- Submit all scripts in a single folder
 - Folder name: `advanced-shell-assignment`
 - Each task must be a separate script file
 - Naming format: `adv_task1.sh`, `adv_task2.sh`, ...
 - All scripts must be executable
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Completion Objective

This assignment is designed to push participants beyond training content and develop **real automation engineering skills**.

Successful completion shows the participant can: - Think independently - Design automation systems - Build logic-based scripts - Solve real system problems - Work like an automation engineer

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